

Ashley J. Chen

+1 (425) 922-7551
ajchen8@illinois.edu

in linkedin.com/in/ashleyjychen
github.com/ash-jyc

RESEARCH OBJECTIVE My research focuses on security frameworks for composable systems, with end-to-end guarantees that persist across complex, dynamic pipelines. My work includes security analysis for vibe coding, jailbreaking online RL coding assistants, authentic media provenance for deepfake prevention, and smart contract verification.

EDUCATION

University of Illinois Urbana-Champaign Aug 2026 – Present
Ph.D. in Computer Science
Advisor: Prof. Luyi Xing

New York University Shanghai Sep 2022 – May 2026
B.S. in Computer Science; Minors in Cybersecurity and Data Science
GPA: 3.79/4.00

- **Relevant Coursework:** Algorithms, Machine Learning, Operating Systems, Software Engineering, Penetration Testing, Application Security, Computer Security, Network Security, Human-AI Alignment

RESEARCH EXPERIENCE

Undergraduate Research Assistant (Capstone) Sep 2025 – Present
NYU Abu Dhabi (eBrain Lab)
Advisor: Prof. Muhammad Shafique

- Analyzed the relationship between security vulnerabilities in LLM code generation and prompting techniques in “vibe coding” for a senior thesis.
- Constructed two datasets of vibe-coding applications: organized a global hackathon with 40 submissions and built a simulation taxonomy to systematically generate vibe coding samples.
- Designed an agentic pipeline to identify CWEs and perform penetration tests on generated apps.

Undergraduate Research Assistant (Independent Study) Oct 2025 – May 2026
NYU Shanghai (BiAlign Lab)
Advisors: Xufang Luo (Microsoft) and Prof. Hua Shen

- Formulated an independent study on online reinforcement learning for coding LLM assistants.
- Trained Qwen with Code-R1, LeetCode, and TACO datasets using RL techniques following DeepSeek-R1 paper.
- Designed a reward model from 1.7k vulnerable prompts and measured vulnerabilities with CodeQL, Semgrep, and Bandit.

Cybersecurity Research Assistant (Cyber NYC REU) May 2025 – Aug 2025
Columbia University (MobileX Lab)
Advisor: Prof. Xia Zhou

- Built a framework for point-of-capture speech security and deepfake prevention, embedding robust physical signatures through echo hiding of real-time transcriptions.
- Created a hash chain by echo hiding a previous three-second window’s hashed transcription in the next window.
- Refactored WhisperFlow (MobiSys ’25) for voice activity detection and Python integration, eliminating silence transcriptions and reducing word error rate by 13%.

Undergraduate Research Assistant Jun 2023 – Aug 2023
Collaborative Project with Microsoft Research
Advisor: Shuo Chen

- Prototyped runtime concolic technology to verify smart-contract transaction safety properties at scale, demonstrated on Uniswap V2.
- Created a Python symbolic-execution engine translating concrete EVM traces into the Boogie verification

language.

- Improved verification efficiency with minimal overhead (0.20% on ERC20 and 0.57% on Uniswap V2), and cut roughly 25% of nonessential bytecode per trace.

PUBLICATIONS
(IN PREPARATION)

Code for All: Educational Applications of the “Vibe Coding” Hackathon in Programming Education across All Skill Levels [[arXiv:2604.22747v1](https://arxiv.org/abs/2604.22747v1)]

Ashley J. Chen, Yijia Cao, Minghao Shao, Ramesh Karri, Muhammad Shafique.

Theorem-Carrying Transactions: Runtime Verification to Ensure Interface Specifications for Smart Contract Safety [[arXiv:2408.06478](https://arxiv.org/abs/2408.06478)]

Thomas Ball, Nikolaj Bjørner, **Ashley J. Chen***, Shuo Chen, Yang Chen, Zhongxin Guo, Tzu-Han Hsu, Peng Liu, Nanqing Luo.

* Authors are listed in alphabetical order.

INDUSTRY
EXPERIENCE

Software Engineer

NYU Blockchain & Fintech – zkSync Research Team

Oct 2024 – May 2025

New York, NY

- Optimized Reclaim Protocol SDK smart contracts for zkSync using zkEVM, reducing gas costs by 10%.
- Researched protocol and encryption differences among MPC, proxy, and TEE zkTLS providers.
- Joined weekly meetings with zkSync and proposed protocol enhancements to support Groth16 ZKPs.

Developer Success Engineering Intern

Mantle Network

Jan 2024 – May 2024

Shanghai, China

- Resolved 100+ developer-support tickets covering Ethereum Layer 2 protocols using Jira.
- Improved cross-functional communication to accelerate ticket resolution and support efficiency.
- Debugged Solidity and Go contracts with cross-functional teams and documented performance metrics.

HONORS
& AWARDS

Founder’s Day Award (Top 40% of Graduates)

NYU Shanghai

May 2026

SURGE Fellowship

UIUC Grainger School of Engineering

Mar 2026

Dean’s List for Academic Year

NYU Shanghai

2023, 2025, 2026

Best Research Project in STEM

NYU Shanghai Undergraduate Research Symposium

Nov 2025

2nd Place & Best Technology Award

NYU Shanghai Digital Innovation Challenge

Dec 2023